

AXIOM

Compound Guide.

The companion to the AXIOM price list. Every compound in the catalogue, pathway by pathway — what it is, how it works, what the research shows, and the context it belongs to.

COMPOUNDS

38

PATHWAYS

8

EVIDENCE

Graded

PRICING

See price list

RESEARCH USE ONLY
IN-VITRO LABORATORY USE
NOT FOR HUMAN OR VETERINARY USE

Compound guide · Edition 1.0
Research catalogue · 38 compounds
No dosing guidance provided

Contents

The companion to the AXIOM price list. Every compound in the catalogue, pathway by pathway: what it is, how it works, what the research actually shows, and the research context it belongs to. Evidence is graded on every card. No pricing here — that lives in the price list.

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EVIDENCE GRADES

- REGISTRATION-GRADE

An approved medicine, or a completed Phase 3 programme behind it.
- HUMAN TRIALS

Controlled human data exists; not approved, or approved only outside the FDA and EMA.
- PRECLINICAL

Cell and animal models only. No completed human trial.

HOW TO READ A COMPOUND ENTRY

Benefit line	What the compound does, in one plain sentence.
Mechanism	The receptor or pathway it acts on.
Route · Acts over	How it is administered, and its documented half-life or action window.
What research shows	Headline findings, with trial figures where they exist.
Safety signals	Known tolerability, or an explicit note where no human safety data exists.
Compared with	How it differs from its nearest neighbour in the same pathway.
Research context	The models it is studied against — context, never a protocol.
Lots · Status	Available lot sizes, and where the compound stands with regulators.

01 GLP-1s & Weight Loss

3 COMPOUNDS

Incretin and amylin biology — the pathway with the strongest human evidence in the catalogue. One approved medicine, one Phase 3 molecule ahead of it in trial results, and one compound that reaches satiety through a different receptor entirely.

Cagrilintide

HUMAN TRIALS

Satiety through the amylin pathway — weight reduction without touching a GLP-1 receptor.

ROUTE

Subcutaneous

ACTS OVER

~180 h half-life

MECHANISM

A long-acting analogue of amylin, the hormone co-secreted with insulin. It activates hindbrain amylin receptors (AMY1R/AMY3R) to raise satiety, slow gastric emptying and suppress post-prandial glucagon — a mechanistically distinct route from the incretin class.⁵

WHAT RESEARCH SHOWS

- **11.8% mean weight reduction** at 68 weeks as monotherapy in a late-stage trial of 3,417 participants; 31.6% lost $\geq$ 15% of body weight vs 4.7% on placebo.⁴
- Paired with semaglutide (CagriSema), reductions deepen beyond either agent alone.
- 2025 mechanism work confirmed the effect runs through central amylin receptors — delete them and the weight loss disappears.⁵

SAFETY SIGNALS

Amylin-class tolerability — early nausea that eases with exposure; the long half-life gives steadier levels.

COMPARED WITH

Unlike the GLP-1 agonists beside it, it works through amylin receptors, so it is studied as a complement to them rather than a substitute.

RESEARCH CONTEXT

The non-incretin satiety arm of weight-management research. Studied where GLP-1 pathways are already saturated, or where a complementary appetite mechanism is the question.

LOTS 10 mg

STATUS Phase 3 (CagriSema program); not yet approved

Retatrutide

HUMAN TRIALS

The strongest weight-loss result yet recorded in clinical trials — nearly 29% of body weight.

ROUTE

Subcutaneous

ACTS OVER

~6-day half-life

MECHANISM

A single molecule agonising three receptors at once — GLP-1, GIP and glucagon. The first two suppress appetite and improve incretin signalling; the glucagon arm raises energy expenditure, which is what separates it from the dual agonists.³

WHAT RESEARCH SHOWS

- Phase 2 obesity: **24.2% mean weight loss** at 48 weeks (top dose) vs 2.1% placebo.
- Phase 3 TRIUMPH-4: up to **28.7% mean weight reduction** at 68 weeks, with meaningful relief of osteoarthritis knee pain.¹
- Liver fat reduced up to 82.4% in a Phase 2a MASLD trial.²

SAFETY SIGNALS

Dose-dependent gastrointestinal effects — nausea, vomiting, diarrhoea — the incretin-class signal, amplified at higher exposure.

COMPARED WITH

Adds a third receptor (glucagon) to Tirzepatide's two, which is what drives the extra energy expenditure and the deeper trial results.

RESEARCH CONTEXT

The deepest metabolic signal of any incretin-class molecule. A 14-trial Phase 3 program (22,000+ participants) is running; not yet an approved medicine anywhere.

LOTS 5 mg · 10 mg · 15 mg · 20 mg · 30 mg · 60 mg

STATUS Phase 3 ongoing (TRIUMPH); not yet approved

Tirzepatide

REGISTRATION-GRADE

The approved benchmark of the class — about a fifth of body weight in registration trials.

ROUTE

Subcutaneous

ACTS OVER

~5-day half-life

MECHANISM

A 39-amino-acid dual agonist of the GIP and GLP-1 receptors — the first of the “twincretin” class. Co-activation suppresses appetite, slows gastric emptying and improves insulin sensitivity.⁶

WHAT RESEARCH SHOWS

- SURMOUNT-1: **20.9% mean weight loss** at 72 weeks (highest dose) vs 3.1% placebo.⁷
- At three years: 22.9% reduction and a 94% lower rate of progression to type 2 diabetes.
- Head-to-head (SURMOUNT-5): 20.2% vs semaglutide's 13.7% at 72 weeks.
- FDA-approved as Mounjaro (diabetes) and Zepbound (weight management).

SAFETY SIGNALS

Mild-to-moderate, dose-dependent nausea, vomiting and diarrhoea; the most thoroughly characterised safety profile in this pathway.

COMPARED WITH

Two receptors to Retatrutide's three and Cagrilintide's amylin route — and, unlike either neighbour, an approved medicine.

RESEARCH CONTEXT

The registration-grade reference of the pathway — the molecule every newer incretin compound is read against.

LOTS 10 mg · 30 mg · 40 mg

STATUS FDA-approved (Mounjaro, Zepbound)

02 GH Secretagogues

5 COMPOUNDS

The growth-hormone axis, approached two ways: replace the hormone directly, or stimulate the body's own pulsatile release through GHRH and ghrelin receptors. The blends exist because the two receptor routes are complementary, not redundant.

CJC-1295 (No DAC) + Ipamorelin

PRECLINICAL

A synchronised growth-hormone pulse driven from two receptors at once.

ROUTE

Subcutaneous

ACTS OVER

CJC ~30 min · ipamorelin ~2 h

MECHANISM

CJC-1295 without DAC (Mod GRF 1-29) is a short-acting GHRH analogue; ipamorelin fires the ghrelin receptor. Together they produce a timed, pulsatile GH release that tracks the axis's natural dynamics rather than flattening them.

WHAT RESEARCH SHOWS

- Each component class raises GH and IGF-1 in human pharmacology studies.
- The combination itself has no controlled human trial — the pairing rests on mechanistic complementarity, not outcome data.

SAFETY SIGNALS

No controlled human trial of the combination; the component classes are generally well tolerated in short human studies.

COMPARED WITH

A stimulate-your-own-pulse pair, versus HGH's direct replacement and tesamorelin's single-receptor GHRH route.

RESEARCH CONTEXT

The standard pairing in secretagogue research wherever pulsatile — rather than sustained — GH signalling is the model under study. See the ipamorelin card for the selective half of the pair.

LOTS 10 mg · 20 mg

STATUS Research-only; no approved product

HGH 191AA (Somatropin)

REGISTRATION-GRADE

Growth hormone itself — the reference molecule for the entire axis.

ROUTE

Subcutaneous

ACTS OVER

Short serum half-life; acts via IGF-1

MECHANISM

Recombinant human growth hormone, 191 amino acids, sequence-identical to the pituitary hormone. Binds GH receptors — chiefly hepatic — driving IGF-1 production, lipolysis, and glucose and lipid handling.⁸

WHAT RESEARCH SHOWS

- Decades of registration-grade trials behind FDA approvals: paediatric growth failure, adult GH deficiency, Turner syndrome, HIV-associated wasting.⁸
- The most completely characterised molecule in this pathway, by a wide margin.

SAFETY SIGNALS

At replacement doses, generally favourable; fluid retention, carpal-tunnel symptoms, joint pain and raised glucose appear dose-dependently and rise sharply at supraphysiologic doses.

COMPARED WITH

The hormone itself, rather than a secretagogue that coaxes the body into releasing more of its own.

RESEARCH CONTEXT

Direct replacement — the ceiling against which every secretagogue below it is measured. Where the model needs the hormone itself rather than a stimulated pulse, this is the tool.

LOTS 36 IU · 40 IU

STATUS FDA-approved (multiple brands)

Ipamorelin

PRECLINICAL

The cleanest pulse in the secretagogue class — GH without cortisol, prolactin or appetite signal.

ROUTE

Subcutaneous

ACTS OVER

~2 h half-life

MECHANISM

A pentapeptide agonist of the ghrelin receptor (GHS-R), and the most selective of its class: it releases GH in discrete pulses without the cortisol, prolactin or hunger side-signals of earlier GHRPs.

WHAT RESEARCH SHOWS

- Human pharmacokinetic studies confirm clean, selective GH pulses.
- No efficacy RCTs; a clinical program in post-operative ileus was discontinued, leaving selectivity — not outcomes — as its documented property.

SAFETY SIGNALS

Clean, selective GH pulses in human pharmacology studies, without the cortisol or prolactin rise of older GHRPs; no efficacy RCT.

COMPARED WITH

The most selective agent of the ghrelin-receptor class — the “clean” half of the standard CJC pairing.

RESEARCH CONTEXT

Chosen wherever selectivity matters — isolating the GH pulse from confounding hormonal noise in axis research.

LOTS 10 mg

STATUS Research-only; ileus program discontinued

Ipamorelin + Tesamorelin

PRECLINICAL

The selective ghrelin pulse paired with the one GHRH analogue that carries registration-grade evidence.

ROUTE

Subcutaneous

ACTS OVER

Ipamorelin ~2 h · tesamorelin minutes

MECHANISM

The same dual-receptor logic as CJC + ipamorelin, with tesamorelin — a clinically approved molecule — as the GHRH arm.

WHAT RESEARCH SHOWS

- Tesamorelin alone holds Phase 3 data (visceral fat –15%; see its card).
- The combination itself is untested in controlled trials.

SAFETY SIGNALS

The tesamorelin arm carries label-grade safety data; the combination itself is untested in controlled trials.

COMPARED WITH

Same pulse-pair logic as the CJC blend, but with a clinically approved molecule (tesamorelin) as the GHRH arm.

RESEARCH CONTEXT

GH-axis research where the GHRH component should be the one with human trial evidence behind it.

LOTS 18 mg

STATUS Research-only combination

Tesamorelin

REGISTRATION-GRADE

The registration-grade GHRH analogue — documented visceral-fat reduction in Phase 3 trials.

ROUTE

Subcutaneous

ACTS OVER

~8–38 min half-life

MECHANISM

A 44-amino-acid GHRH analogue. Binds pituitary GHRH receptors, restoring pulsatile GH secretion and raising IGF-1; the downstream shift in adipose metabolism runs specifically against visceral fat.

WHAT RESEARCH SHOWS

- Two Phase 3 trials (pooled n = 806): visceral adipose tissue reduced **about 15%** vs placebo over 26 weeks, with liver-fat reductions reported.
- FDA-approved in 2010 as Egrifta for HIV-associated lipodystrophy — the trials were in that population; use beyond it is off-label territory.

SAFETY SIGNALS

Injection-site reactions, joint pain, fluid retention and a small upward drift in fasting glucose — documented across its Phase 3 trials.

COMPARED WITH

The one GHRH analogue here with registration-grade human evidence behind it.

RESEARCH CONTEXT

Visceral-adiposity and GH-axis restoration models — the only GHRH analogue with human registration evidence.

LOTS 10 mg · 20 mg

STATUS FDA-approved (Egrifta) for HIV lipodystrophy

03 Healing & Repair

8 COMPOUNDS

Tissue repair decomposed into four mechanisms — angiogenesis, cell migration, matrix remodelling, inflammation control — each with its own compound, plus the blends that combine them. Evidence here is mostly preclinical, with two human - tested exceptions.

BPC-157

PRECLINICAL

The most-studied repair peptide in the catalogue — soft tissue, tendon and gut lining in animal models.

ROUTE

Subcutaneous (orally stable)

ACTS OVER

Rodent half-life ~15 min; effects outlast it

MECHANISM

A synthetic 15-amino-acid fragment of a gastric body-protection compound. Upregulates VEGF-driven angiogenesis and promotes fibroblast migration; raises growth-hormone-receptor expression in tendon cells.¹⁰

WHAT RESEARCH SHOWS

- Accelerated healing of transected Achilles tendon, muscle, ligament and gut lesions across a large rodent literature.⁹
- Consistent gut-protective effects in NSAID-damage models.
- No completed human RCT — a 2025 systematic review found the clinical evidence gap unclosed; prohibited in professional sport.¹¹

SAFETY SIGNALS

No human safety data — the record is rodent and anecdotal; prohibited in professional sport.

COMPARED WITH

The angiogenic (VEGF) base of the repair blends, complementary to TB-500's cell-migration route.

RESEARCH CONTEXT

Soft-tissue, tendon and gut-lining injury models — the angiogenic arm of repair research, and the base layer of both repair blends.

LOTS 10 mg

STATUS Research-only; no completed human trial

BPC-157 + TB-500 (Wolverine)

PRECLINICAL

Angiogenesis and cell migration in one preparation — the two repair axes that don't overlap.

ROUTE

Subcutaneous

ACTS OVER

Component half-lives (minutes to ~2 h)

MECHANISM

BPC-157 supplies the vascular signal (VEGF, angiogenesis); TB-500 supplies the cellular one (actin-mediated migration to the injury site). The mechanisms are complementary, which is the blend's entire rationale.

WHAT RESEARCH SHOWS

- Each component carries a deep rodent literature (see their cards).
- The blend itself has no published study — it is a composition, not a compound.

SAFETY SIGNALS

No human safety data for either component or the blend.

COMPARED WITH

Combines the two flagship repair mechanisms — angiogenesis and cell migration — that its single-agent cards cover separately.

RESEARCH CONTEXT

Combined soft-tissue repair models where both the vascular and the migratory mechanism are wanted in one protocol design.

LOTS 20 mg

STATUS Research-only composition

GHK-Cu

HUMAN TRIALS

Collagen remodelling with human trial data behind it.

ROUTE

Topical or subcutaneous

ACTS OVER

Active at nanomolar levels

MECHANISM

A naturally occurring copper-binding tripeptide (glycyl-histidyl-lysine). At nanomolar concentrations it modulates collagen synthesis and breakdown, metalloproteinase activity, angiogenesis and antioxidant enzymes — a remodelling regulator rather than a blunt growth signal.¹²

WHAT RESEARCH SHOWS

- Human topical trial: collagen increase in **70% of volunteers**, outperforming vitamin C and retinoic acid.¹²
- 2024 RCT (n = 60): 31% wrinkle reduction and 28% elasticity gain over 12 weeks vs placebo.
- Animal wound models: faster closure, more vessel formation, higher antioxidant enzyme levels.

SAFETY SIGNALS

Topical use has a strong cosmetic-grade safety record; injectable use lacks long-term data and carries a theoretical copper-load risk.

COMPARED WITH

The matrix-remodelling arm of the repair set — a regulator, not a blunt growth signal like the growth-factor cards.

RESEARCH CONTEXT

Skin-remodelling and wound-repair research. The evidence base is cosmetic-grade human data — real, but no drug approval behind it.

LOTS 50 mg · 100 mg

STATUS Cosmetic ingredient; no drug approval

KPV

PRECLINICAL

The anti-inflammatory signal of alpha-MSH, without its pigmentary activity.

ROUTE

Subcutaneous, oral or topical

ACTS OVER

No measured human pharmacokinetics

MECHANISM

The C-terminal tripeptide of alpha-MSH (lysine-proline-valine). It keeps the parent hormone's suppression of NF-κB-driven cytokines — TNF-α, IL-6 — while shedding the melanocortin pigmentary signalling.¹³

WHAT RESEARCH SHOWS

- Significant anti-inflammatory effect in two murine colitis models: earlier recovery, reduced inflammatory infiltrates and MPO activity.¹³
- Supporting in-vitro work on intestinal barrier protection under cytokine stress.

SAFETY SIGNALS

No human safety data; the evidence is murine colitis models.

COMPARED WITH

The inflammation-control arm of the KLOW blend, distinct from the growth and migration signals around it.

RESEARCH CONTEXT

Gut-inflammation and barrier-integrity models; the inflammation-control arm of the KLOW blend.

LOTS 10 mg

STATUS Research-only

KLOW (BPC-157 + TB-500 + GHK-Cu + KPV)

PRECLINICAL

Four repair mechanisms in a single preparation.

ROUTE

Subcutaneous

ACTS OVER

Four component half-lives

MECHANISM

The full decomposition of tissue repair, stacked: angiogenesis (BPC-157), cell migration (TB-500), matrix remodelling (GHK-Cu) and inflammation control (KPV). Each arm's mechanism is documented on its own card.

WHAT RESEARCH SHOWS

- No study of the four-way combination exists; the evidence is entirely the sum of its parts.
- GHK-Cu is the only arm with human data.

SAFETY SIGNALS

No human data for the combination; only the GHK-Cu arm has human evidence.

COMPARED WITH

Stacks all four repair mechanisms at once — the broadest of the blends, and a composition rather than a studied compound.

RESEARCH CONTEXT

Broad-spectrum tissue-repair models where all four mechanisms are wanted simultaneously and component-level attribution is not the question.

LOTS 80 mg

STATUS Research-only composition

LL-37

HUMAN TRIALS

The body's own antimicrobial — wound closure where healing has stalled.

ROUTE

Topical or local

ACTS OVER

Resistant to proteolytic breakdown

MECHANISM

The only cathelicidin encoded in the human genome: a 37-amino-acid peptide stored in neutrophil granules. Disrupts microbial membranes and biofilms while signalling re-epithelialisation, granulation and angiogenesis at the wound site.¹⁴

WHAT RESEARCH SHOWS

- Topical LL-37 increased re-epithelialisation, granulation, collagen organisation and angiogenesis in wound models.¹⁵
- An early-phase clinical trial in hard-to-heal ulcers reported complete closure in roughly 28% of the large-wound group versus 8% on placebo.
- Broad activity against Gram-positive and Gram-negative pathogens, MRSA models included.

SAFETY SIGNALS

Cytotoxic to host cells at high concentrations; early-phase wound trials were tolerated.

COMPARED WITH

The antimicrobial arm of repair — where healing meets innate immunity, unlike the purely regenerative cards.

RESEARCH CONTEXT

Infected and stalled wound models — the point where repair research meets innate immunity.

LOTS 5 mg

STATUS No approval; early-phase trials only

Peg MGF

PRECLINICAL

The repair signal of loaded muscle, extended to a research-usable half-life.

ROUTE

Subcutaneous

ACTS OVER

PEGylation extends half-life to ~24 h

MECHANISM

The pegylated E-domain of mechano growth factor — the IGF-1 splice variant muscle produces under mechanical load. Proposed to activate muscle satellite cells; PEGylation stretches the half-life from minutes to around 24 hours.

WHAT RESEARCH SHOWS

- Satellite-cell activation in rodent and in-vitro work — largely from one research group; an independent laboratory failed to reproduce the myoblast effect.
- Endogenous MGF is confirmed exercise-responsive in human muscle biopsies, but no injected-MGF human study exists.

SAFETY SIGNALS

No human safety data, and a documented replication gap in the efficacy work.

COMPARED WITH

The muscle-repair signalling arm — the weakest-replicated card in this pathway.

RESEARCH CONTEXT

Muscle-repair signalling models. The weakest-replicated compound in this pathway — read it accordingly.

LOTS 2 mg

STATUS Research-only

TB-500

PRECLINICAL

Cell migration to the injury site — the actin-binding arm of repair research.

ROUTE

Subcutaneous

ACTS OVER

~2 h half-life

MECHANISM

A synthetic fragment of thymosin beta-4 covering its actin-binding segment. Sequesters G-actin and drives cell migration through ILK/Akt signalling — keratinocyte migration rises about 42% in scratch assays — with downstream angiogenesis.¹⁶

WHAT RESEARCH SHOWS

- A 2004 *Nature* study established the parent peptide's cardioprotection after infarction in mice.¹⁶
- The parent reached human trials (RGN-352 cardiac; RGN-259 ophthalmic); TB-500 itself has no completed human data.

SAFETY SIGNALS

No human safety data of its own; the parent peptide reached human trials as RGN-352 and RGN-259.

COMPARED WITH

The cell-migration arm, paired with BPC-157's angiogenesis in the Wolverine blend.

RESEARCH CONTEXT

The cell-migration mechanism across wound, cardiac and corneal repair models; the migratory half of the Wolverine blend.

LOTS 10 mg

STATUS Research-only; parent reached trials

04 Brain Health & Nootropics

6 COMPOUNDS

Neurotrophic and regulatory peptides, most of them from the Russian clinical tradition — which means real human data for some, and honest gaps for others. The spread of evidence grades in this pathway is the widest in the catalogue.

Adamax

PRECLINICAL

A designed successor to Semax — with no published evidence of its own yet.

ROUTE

Intranasal

ACTS OVER

No measured pharmacokinetics

MECHANISM

Semax modified with an adamantane cage on the C-terminus — chemistry borrowed from CNS drugs like memantine — intended to extend half-life and central penetration. Its mechanism is inferred from the parent, not measured.

WHAT RESEARCH SHOWS

- No published study in any species, in vitro or in vivo. Every specific claim about it traces to manufacturer inference from Semax.
- What is real is the parent molecule's record — see the Semax card.

SAFETY SIGNALS

No safety data in any species — every property is inferred from Semax.

COMPARED WITH

A theoretical “Semax-plus” built for longer action; the parent (Semax) is the one with the real record.

RESEARCH CONTEXT

For researchers who want the derivative chemistry specifically. Stated plainly: this card carries the least evidence in the catalogue.

LOTS 10 mg

STATUS Research-only; no published studies

Cerebrolysin

HUMAN TRIALS

The most clinically documented neurotrophic in the catalogue — stroke and dementia trial data.

ROUTE

Intravenous or intramuscular

ACTS OVER

Peptide mixture; infused slowly

MECHANISM

A mixture of low-molecular-weight neurotrophic peptides that mimics endogenous growth factors and induces more of them — BDNF, NGF, IGF-1 and VEGF upregulation are documented.¹⁷

WHAT RESEARCH SHOWS

- Meta-analysis of 14 RCTs in acute ischaemic stroke: significantly better neurological recovery (NIHSS +1.39), with no excess of serious adverse events.¹⁸
- A 24-week multicentre RCT reported improved outcomes in vascular dementia; reviews suggest small benefits in Alzheimer's symptoms.
- A prescription drug in dozens of countries; not FDA-approved.

SAFETY SIGNALS

Generally mild and transient — dizziness, injection-site reactions, transient temperature rise; rare anaphylaxis, as it is animal-tissue-derived, and infusion-rate-related cardiovascular effects.

COMPARED WITH

The most thoroughly evidenced neurotrophic in this pathway, with meta-analysed trial data the others lack.

RESEARCH CONTEXT

Neurorecovery models — post-stroke and cognitive-decline research where a clinically characterised neurotrophic is required.

LOTS 60 mg

STATUS Prescription drug in dozens of countries; not FDA

Pinealon

PRECLINICAL

A gene-level neuronal bioregulator hypothesis — cell and rodent data only.

ROUTE

Subcutaneous or intranasal

ACTS OVER

No measured human pharmacokinetics

MECHANISM

The Khavinson tripeptide EDR (Glu-Asp-Arg). Hypothesised to cross into the neuronal nucleus and modulate gene expression directly, with antioxidant and anti-apoptotic signals reported in cell and rodent work.

WHAT RESEARCH SHOWS

- The entire evidence base is in-vitro and rodent, produced within one affiliated research network.
- No completed human RCT and no independent Western replication.

SAFETY SIGNALS

No human safety data; the base is cell and rodent work from one network.

COMPARED WITH

The neuronal counterpart to Epithalon in pathway 08 — same bioregulator program, different tissue target.

RESEARCH CONTEXT

Neuronal oxidative-stress models inside the peptide-bioregulator program; paired conceptually with Epithalon in pathway 08.

LOTS 20 mg

STATUS Research-only; single research network

Selank

HUMAN TRIALS

Anxiety reduction comparable to benzodiazepines in Russian trials — without the sedation.

ROUTE

Intranasal

ACTS OVER

Short half-life; multiple daily use

MECHANISM

A heptapeptide analogue of tuftsin, an immunoglobulin fragment. It does not bind GABA-A directly; instead it modulates GABAergic and serotonergic tone and raises BDNF expression.

WHAT RESEARCH SHOWS

- Russian trials in generalised anxiety and asthenia reported anxiety reduction broadly comparable to benzodiazepine comparators, with a mild activating rather than sedating profile.
- Approved in Russia (2009) for generalised anxiety disorder; not FDA or EMA recognised.

SAFETY SIGNALS

Well tolerated across its Russian clinical use; the common complaint is mild nasal irritation.

COMPARED WITH

The anxiolytic half of the pair — calm without the sedation of a benzodiazepine or the activation of Semax.

RESEARCH CONTEXT

The anxiolytic arm of the Russian pair — anxiety-with-fatigue models where sedation is a confound to avoid.

LOTS 10 mg

STATUS Approved in Russia (2009); not FDA/EMA

Semax

HUMAN TRIALS

Two decades of clinical use in stroke recovery and cognition.

ROUTE

Intranasal (or subcutaneous)

ACTS OVER

~30 min for active metabolites

MECHANISM

An ACTH-derived heptapeptide stripped of hormonal activity. Signals through melanocortin receptors, upregulating BDNF and NGF in hippocampus and cortex and suppressing inflammatory gene expression after ischaemia.¹⁹

WHAT RESEARCH SHOWS

- On Russia's essential-medicines list; more than twenty years of institutional use in stroke recovery, optic-nerve disease and cognitive impairment.¹⁹
- A 2018 study of 110 post-stroke patients found plasma BDNF rose alongside motor and independence gains — open-label, so directional rather than definitive.

SAFETY SIGNALS

Well tolerated over two decades of Russian use; mild nasal irritation is the usual report.

COMPARED WITH

The cognitive and neuroprotective half of the pair, alongside Selank's anxiolytic action.

RESEARCH CONTEXT

The cognitive and neuroprotective arm of the pair — stroke-recovery and attention models.

LOTS 10 mg

STATUS Russia essential-medicines list; not FDA/EMA

Selank + Semax

HUMAN TRIALS

Calm and focus studied side by side — the two Russian regulatory peptides together.

ROUTE

Intranasal

ACTS OVER

Short component half-lives

MECHANISM

Complementary central actions: Selank's GABA/serotonin-modulated anxiolysis alongside Semax's BDNF/NGF-driven cognitive signal. Both converge on raised BDNF from different directions.

WHAT RESEARCH SHOWS

- Each component is clinically documented in Russia (see their cards); the tier reflects the components.
- The blend itself is a research composition without its own trial.

SAFETY SIGNALS

Each component is well tolerated in Russian use; the blend itself is a research composition.

COMPARED WITH

Puts the anxiolytic (Selank) and nootropic (Semax) questions together rather than testing them apart.

RESEARCH CONTEXT

Models where the anxiolytic and nootropic questions are asked together rather than separately.

LOTS 20 mg

STATUS Components approved in Russia; blend research-only

05 Energy & Endurance

5 COMPOUNDS

Metabolic switches — compounds that engage the pathways exercise normally triggers: AMPK, ERR, mitochondrial biogenesis, fat oxidation. The class is defined by rich rodent evidence and, so far, sparse human data. Several are WADA - prohibited exercise mimetics.

5-Amino-1MQ

PRECLINICAL

An NAD⁺-preserving switch aimed specifically at fat tissue.

ROUTE
Oral

ACTS OVER
~4–7 h half-life (rodent)

MECHANISM

A small-molecule NNMT inhibitor (not a peptide). Blocking NNMT stops it consuming nicotinamide, preserving it for the NAD⁺ salvage pathway — and NNMT sits highest in adipose tissue, which gives the mechanism a fat-selective rationale.²⁰

WHAT RESEARCH SHOWS

- Diet-induced obese mice lost body weight and white fat with no change in food intake.²⁰
- A 2024 follow-up: fat-mass gain suppressed roughly 72% with lean mass unchanged, plus better glucose tolerance and less fatty liver.
- No completed human trial; the mouse data is real and replicated, human translation is not established.

SAFETY SIGNALS

Very limited human safety data — the evidence is mouse studies.

COMPARED WITH

An NAD⁺-preserving route to fat metabolism, distinct from AICAR's and SLU-PP-332's direct exercise-signal switches.

RESEARCH CONTEXT

Adipose-targeted NAD⁺ metabolism models.

LOTS 50 mg STATUS Research-only; not approved

AICAR

PRECLINICAL

The original exercise mimetic — endurance gains in untrained mice.

ROUTE
Intravenous or subcutaneous

ACTS OVER
~2–3 h half-life

MECHANISM

An AMP-mimetic nucleoside that directly activates AMPK, switching on the fasted-exercise transcriptional program: fat oxidation and mitochondrial biogenesis without the workout.²¹

WHAT RESEARCH SHOWS

- The landmark 2008 *Cell* study: **+44% running endurance** in sedentary mice with no training, via the AMPK-PPAR δ pathway.²¹
- WADA-prohibited as a metabolic modulator — the first exercise mimetic to be banned.
- No published human endurance data.

SAFETY SIGNALS

No human endurance safety data; WADA-prohibited as a metabolic modulator.

COMPARED WITH

The AMPK route to the exercise program, versus SLU-PP-332's ERR route to the same end.

RESEARCH CONTEXT

AMPK-pathway energetics models — the reference exercise mimetic, paired conceptually with SLU-PP-332's ERR route.

LOTS 50 mg STATUS Research-only; WADA-prohibited

L-Carnitine (Injectable)

REGISTRATION-GRADE

A fat-oxidation shuttle, delivered at full systemic dose.

ROUTE

Intravenous or intramuscular

ACTS OVER

~10–23 h half-life

MECHANISM

A conditionally essential amino-acid derivative that carries long-chain fatty acids into mitochondria for beta-oxidation. The injectable route bypasses oral bioavailability of only 14–18%, delivering essentially the full dose systemically.²²

WHAT RESEARCH SHOWS

- Systematic reviews: improved high-intensity performance at 3–4 g pre-exercise or 2–2.72 g/day chronically; no effect at moderate intensity.²²
- A nine-month RCT found greater fat oxidation when combined with aerobic training.
- The molecule is a long-established clinical agent; evidence for the performance use is graded modest.

SAFETY SIGNALS

No side effects attributable to carnitine in intravenous infusion studies; well tolerated.

COMPARED WITH

A substrate shuttle rather than a signalling switch — it moves fat into mitochondria, it doesn't reprogram them.

RESEARCH CONTEXT

Fat-oxidation and recovery models where complete systemic delivery is the variable of interest.

LOTS 5000 mg

STATUS Approved molecule (Carnitor)

MOTS-c

PRECLINICAL

A mitochondrial signal the body raises during exercise — and loses with age.

ROUTE

Subcutaneous

ACTS OVER

~1–4 h half-life (rodent)

MECHANISM

A 16-amino-acid peptide encoded within mitochondrial 12S rRNA. Acting through AMPK and PGC-1 α , it regulates glucose use and fat oxidation, translocating to the nucleus under metabolic stress.²³

WHAT RESEARCH SHOWS

- Exercise raises endogenous MOTS-c in human muscle and blood; levels decline with age.²³
- Restored insulin sensitivity in aged mice; 2025 work reported improved intrinsic muscle mitochondrial bioenergetics.
- Exogenous human trials not completed.

SAFETY SIGNALS

No human safety data; endogenous exercise-response biology is well documented, exogenous dosing is not.

COMPARED WITH

A mitochondrial exercise-signal, sharing its origin with Humanin in pathway 08.

RESEARCH CONTEXT

Mitochondrial “exercise-signal” models and metabolism-with-age research. Shares a mitochondrial origin with Humanin in pathway 08.

LOTS 10 mg · 40 mg

STATUS Research-only

SLU-PP-332 (Injectable)

PRECLINICAL

The newest exercise-mimetic axis — endurance biology through the ERR receptors.

ROUTE

Subcutaneous

ACTS OVER

No measured human pharmacokinetics

MECHANISM

A small-molecule pan-agonist of the estrogen-related receptors $ERR\alpha/\beta/\gamma$. It drives the endurance-exercise transcriptional program — mitochondrial biogenesis and fatty-acid-oxidation genes — and shifts muscle toward oxidative type IIa fibres.²⁴

WHAT RESEARCH SHOWS

- A 2023 study reported increased treadmill endurance in mice, with reduced fat mass and improved insulin sensitivity in follow-ups.²⁴
- Mouse-only to date; WADA-classified as an exercise mimetic.

SAFETY SIGNALS

No human safety data; mouse studies reported no severe adverse events; WADA-classified exercise mimetic.

COMPARED WITH

The ERR route to the endurance program, a different receptor family from AICAR's AMPK.

RESEARCH CONTEXT

The ERR arm of exercise-mimetic research — a distinct receptor route from AICAR's AMPK.

LOTS 5 mg

STATUS Research-only; mouse-only data

06 Immunity

1 COMPOUND

A single, well-evidenced entry — the immune-modulating thymic peptide approved as a medicine across dozens of countries.

Thymosin Alpha-1

HUMAN TRIALS

An immune-system modulator with controlled trial data in infection and sepsis.

ROUTE

Subcutaneous

ACTS OVER

~2 h half-life

MECHANISM

A 28-amino-acid thymic peptide (thymalfasin). Stimulates T-cell maturation and interferon- γ secretion, broadening both innate and adaptive immune responses.²⁵

WHAT RESEARCH SHOWS

- Approved in around 37 countries (as Zadaxin) — in China for hepatitis B and vaccine enhancement; not FDA-approved.
- Sepsis meta-analysis: reduced mortality (RR 0.59); the TESTS trial (n = 1,106) is the strongest controlled evidence to date.²⁵
- Decades of clinical safety record.

SAFETY SIGNALS

An excellent safety record — mild injection-site discomfort is the main report; no significant adverse reactions across 2,000+ treated patients.

COMPARED WITH

The single immunity entry — an immune modulator rather than a repair or metabolic agent.

RESEARCH CONTEXT

Immune-restoration models — chronic viral infection, sepsis adjunct, and immunosenescence research.

LOTS 10 mg

STATUS Approved in ~37 countries (Zadaxin); not FDA

07 Sexual Health

5 COMPOUNDS

The hypothalamic-pituitary-gonadal axis approached at every level — upstream at the hypothalamus (kisspeptin), directly at the gonadal receptor (HCG, HMG), and centrally in the desire circuits of the brain (PT-141). Three registration-grade medicines sit in this pathway.

HCG

REGISTRATION-GRADE

An LH mimic — the standard tool for driving gonadal testosterone.

ROUTE

Subcutaneous or intramuscular

ACTS OVER

Long half-life; sustained LH-like action

MECHANISM

A placental glycoprotein hormone that binds the shared LH/hCG receptor, mimicking luteinising hormone: Leydig-cell testosterone production in men, ovulation trigger in women.²⁶

WHAT RESEARCH SHOWS

- Standard of care in hypogonadotropic hypogonadism and assisted reproduction.²⁶
- Maintains intratesticular testosterone during exogenous-testosterone suppression — one of its most common clinical uses.

SAFETY SIGNALS

Well-established clinical agent; the main risk in stimulation settings is ovarian hyperstimulation, plus injection-site reactions.

COMPARED WITH

Acts directly at the gonadal LH receptor, where Kisspeptin acts one level upstream at the hypothalamus.

RESEARCH CONTEXT

HPG-axis maintenance and fertility models — the direct-receptor route.

LOTS

10000 IU

STATUS

Approved medicine

HMG

REGISTRATION-GRADE

Both gonadotropins in one preparation — the FSH complement to HCG.

ROUTE

Subcutaneous or intramuscular

ACTS OVER

~32 h half-life

MECHANISM

Human menopausal gonadotropin (menotropin) carries both FSH and LH activity: folliculogenesis in women, spermatogenesis induction in men.²⁷

WHAT RESEARCH SHOWS

- Registration-grade use in IVF ovarian stimulation, with efficacy comparable to recombinant FSH.²⁷
- Induces spermatogenesis in hypogonadotropic hypogonadism where testicular function is intact.

SAFETY SIGNALS

Ovarian hyperstimulation and multiple-pregnancy risk in stimulation settings; abdominal and injection-site effects.

COMPARED WITH

Supplies both FSH and LH, where HCG supplies LH-like action alone.

RESEARCH CONTEXT

The FSH side of full-axis fertility research — the complement to HCG's LH-only action.

LOTS

75 IU

STATUS

Approved medicine (Menopur)

07 · SEXUAL HEALTH

19 OF 28

Kisspeptin

HUMAN TRIALS

The master switch at the top of the reproductive axis.

ROUTE

Subcutaneous, IV or intranasal

ACTS OVER

~4–28 min half-life

MECHANISM

A hypothalamic neuropeptide (KISS1 gene) that triggers GnRH neurons — sitting one level upstream of the pituitary, so it drives the body's own pulsatile LH rather than substituting for it.²⁸

WHAT RESEARCH SHOWS

- Human studies: dose-dependent LH rise and increased pulse frequency.²⁸
- Raised testosterone in hypotestosteronaemic men with type 2 diabetes in a proof-of-concept study.²⁹

SAFETY SIGNALS

Human trials report gonadotropin stimulation without side effects or adverse events.

COMPARED WITH

The upstream axis trigger — it drives the body's own LH pulse, where HCG substitutes for LH directly.

RESEARCH CONTEXT

Endogenous-axis stimulation research — the upstream alternative to HCG's direct receptor agonism.

LOTS 10 mg

STATUS Investigational; IVF-trigger studies

Oxytocin Acetate

HUMAN TRIALS

The bonding hormone — a large but contested social-behaviour literature.

ROUTE

Intranasal, IV or intramuscular

ACTS OVER

~10–15 min half-life

MECHANISM

The endogenous nonapeptide hormone as its acetate salt. Signals through oxytocin receptors; its central roles in bonding, trust and sexual response are studied mostly via intranasal delivery.³⁰

WHAT RESEARCH SHOWS

- A large intranasal literature on social cognition — results are mixed and reproducibility is actively debated.³⁰
- Small multi-week studies suggest signals in some psychiatric populations.
- The molecule is separately an approved obstetric drug.

SAFETY SIGNALS

Intranasal use is generally tolerated in social-cognition studies; the molecule is separately an approved obstetric drug.

COMPARED WITH

The softest evidence in this pathway, and the only entry whose primary study route is behavioural rather than endocrine.

RESEARCH CONTEXT

Social and affective-behaviour research. Read with the field's own caution — the softest evidence in this pathway outside its obstetric indication.

LOTS 10 mg

STATUS Approved obstetric drug; behavioural use investigational

08 Longevity & Cellular Repair

5 COMPOUNDS

Cellular - maintenance biology — sleep architecture, telomeres, mitochondrial signalling, the NAD+ pool. The evidence runs from one FDA - approved mitochondrial medicine down to in - vitro telomere work; each card says where its compound sits.

DSIP

HUMAN TRIALS

A sleep-pressure modulator — most active exactly when sleep is disturbed.

ROUTE

Subcutaneous or intravenous

ACTS OVER

~15–25 min half - life; effects last hours

MECHANISM

Delta sleep - inducing peptide, a nonapeptide named for the delta - wave EEG activity it increases. It behaves as a modulator of sleep pressure rather than a sedative.³²

WHAT RESEARCH SHOWS

- Small double - blind studies in insomniacs: better sleep efficiency and shorter onset latency — effects real but modest.³²
- Most active where sleep is already disrupted; minimal effect in healthy sleepers.³³

SAFETY SIGNALS

An excellent safety record across decades of research; mild drowsiness, brief post - injection nausea and injection - site irritation.

COMPARED WITH

The sleep - architecture entry of the longevity set — a modulator, not a sedative.

RESEARCH CONTEXT

Sleep - architecture and recovery - under - stress models.

LOTS

5 mg · 10 mg

STATUS

Research - only; no approval

Epithalon

PRECLINICAL

Telomerase activation — demonstrated in human cells, not yet in humans.

ROUTE

Subcutaneous or intranasal

ACTS OVER

Short half - life

MECHANISM

The Khavinson tetrapeptide AEDG, a synthetic analogue of the pineal extract epithalamin. Proposed to activate telomerase and support pineal/circadian signalling.³⁴

WHAT RESEARCH SHOWS

- Telomerase activation and telomere elongation in human cell culture — independently replicated at Brunel University in 2025.³⁴
- Rodent lifespan data mixed and regimen - dependent; human evidence is only observational, and with the parent extract rather than the synthetic peptide.

SAFETY SIGNALS

No controlled human safety trial of the synthetic peptide; the parent extract has long observational use.

COMPARED WITH

The telomere - biology counterpart to Pinealon's neuronal work — same Khavinson program.

RESEARCH CONTEXT

Telomere - biology and pineal - aging models; the longevity counterpart to Pinealon's neuronal work.

LOTS

50 mg

STATUS

Research - only

Humanin

PRECLINICAL

A mitochondrial cytoprotectant that tracks with human longevity.

ROUTE

Subcutaneous or intraperitoneal

ACTS OVER

Wild-type ~30 min; analogues > 4 h

MECHANISM

A 24-amino-acid mitochondrial-derived peptide encoded in the 16S rRNA region, with neuroprotective and cytoprotective signalling. Its levels fall with age — but stay high in centenarians' offspring and in negligible-senescence species.³⁵

WHAT RESEARCH SHOWS

- Overexpression extends lifespan in *C. elegans*, dependent on the daf-16/FOXO pathway.³⁵
- The potent analogue HNG improved metabolic healthspan and lowered inflammatory markers in middle-aged mice.

SAFETY SIGNALS

Limited human safety data; a theoretical caution that it may protect cancer cells (BAX interaction) and affect IGF-1 signalling.

COMPARED WITH

Shares its mitochondrial-peptide origin with MOTS-c in pathway 05, on the cytoprotective rather than metabolic side.

RESEARCH CONTEXT

Mitochondrial longevity-signalling and healthspan models; shares its mitochondrial origin with MOTS-c in pathway 05.

LOTS 10 mg

STATUS Research-only; observational human data only

NAD+

HUMAN TRIALS

Refilling the coenzyme pool that halves between your 20s and 50s.

ROUTE

Intravenous (or subcutaneous)

ACTS OVER

The coenzyme itself

MECHANISM

The coenzyme itself, for direct administration — central to energy production, DNA repair and sirtuin signalling. The rationale is restoring a pool that declines roughly 50% across mid-life.³⁷

WHAT RESEARCH SHOWS

- Precursor trials (NR/NMN) reliably raise blood NAD+ and are well tolerated, though functional endpoints remain unclear.³⁷
- A pilot study characterised the plasma fate of a 6-hour IV NAD+ infusion; direct-infusion outcome data is still scarce.³⁶

SAFETY SIGNALS

Direct IV infusion can cause nausea, cramping, chest tightness and flushing tied to infusion rate; precursors (NR, NMN) are well tolerated in trials.

COMPARED WITH

The coenzyme delivered directly, versus the precursor supplements (NR, NMN) that raise it indirectly.

RESEARCH CONTEXT

Cellular-energetics and DNA-repair research. An honest read: robust biochemistry, thin human outcome data.

LOTS 500 mg · 1000 mg

STATUS Precursors trial-backed; direct infusion widely used, thin trial data

Glossary

The recurring biology, in plain terms — enough to read the cards above without a pharmacology background. Definitions describe the mechanism, not any compound's effect.

GLP-1

Glucagon-like peptide-1. An **incretin** gut hormone that raises insulin after eating and signals fullness. The target of the widely-used weight-loss drugs.

GIP

Glucose-dependent insulintropic polypeptide. A second incretin hormone; agonising it alongside GLP-1 is what makes tirzepatide a “dual” agonist.

Glucagon receptor

The receptor for glucagon, a hormone that raises blood sugar and energy expenditure. Adding it to GLP-1 and GIP makes a “triple” agonist.

Incretin

A class of gut hormones released after eating that amplify insulin release. GLP-1 and GIP are the two main ones.

Amylin

A hormone co-released with insulin that promotes fullness and slows stomach emptying — a satiety pathway separate from the incretins.

GHRH

Growth-hormone-releasing hormone. The hypothalamic signal that tells the pituitary to release growth hormone; several compounds here are GHRH analogues.

Ghrelin receptor (GHS-R)

The receptor for ghrelin, the “hunger hormone”, which also triggers growth-hormone release. Ipamorelin acts here.

IGF-1

Insulin-like growth factor 1. The downstream mediator of most growth-hormone effects, produced chiefly by the liver.

HPG axis

The hypothalamic-pituitary-gonadal axis — the hormone chain (GnRH → LH/FSH → testosterone/oestrogen) that governs reproduction.

LH / FSH / GnRH

Luteinising hormone and follicle-stimulating hormone, the two pituitary gonadotropins, released in response to gonadotropin-releasing hormone from the hypothalamus.

Melanocortin receptors

A receptor family (MC1R–MC5R) involved in pigmentation, inflammation and, for MC3R/MC4R, sexual desire. Targeted by PT-141 and KPV.

AMPK

AMP-activated protein kinase — the cell’s “low-energy” sensor. Activating it switches on fat-burning and mitochondrial genes, mimicking exercise.

ERR

Estrogen-related receptors ($\alpha/\beta/\gamma$). Nuclear receptors that drive mitochondrial and endurance genes; the target of SLU-PP-332.

PGC-1 α

The master regulator of mitochondrial biogenesis — the switch that builds more mitochondria in response to exercise signals.

VEGF

Vascular endothelial growth factor. Drives angiogenesis — the growth of new blood vessels — central to wound and tissue repair.

BDNF / NGF

Brain-derived and nerve growth factors. Neurotrophins that support neuron survival, growth and plasticity; several nootropic peptides raise them.

Telomerase

The enzyme that rebuilds telomeres, the protective caps on chromosomes that shorten with each cell division. A longevity research target.

Cardiolipin

A lipid unique to the inner mitochondrial membrane, essential to energy production. SS-31 binds and stabilises it.

NAD⁺

Nicotinamide adenine dinucleotide. A coenzyme central to energy production and DNA repair; its levels fall with age.

Satellite cells

Muscle stem cells that repair and grow muscle fibres after loading or injury. The proposed target of mechano growth factor.

RCT / Phase 2 / Phase 3

A randomised controlled trial is the gold standard for evidence. Phase 2 tests efficacy and dose; Phase 3 confirms it in large populations before approval.

Half-life

The time for half a dose to clear the bloodstream — a measure of how long a compound acts, not a dosing instruction.

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Compliance

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